

HARDWARE STRATEGY · CONFIDENTIAL

HearMe

Hardware & IoT Integration Strategy

VOLUME III

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Product: HearMe — Women's Personal Safety App

Scope: Wearables · Vehicle · Home · Network · B2B Kits

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1. Strategic Framing

Women's safety hardware in India fails on three common axes: **too ugly to wear**, **too obvious to an attacker**, or **too expensive for the mass market**. The HearMe hardware stack is built around three price tiers, each solving a different job. The HearMe app becomes the universal backend they all talk to.

Tier	Price Band	Audience	Job To Be Done
T1 Budget	₹300 – ₹700	Mass market, NGOs, rural	"A panic button I can carry in my pocket."
T2 Fashion	₹800 – ₹3,000	Urban working women, students	"Safety I'll actually wear daily."
T3 Specialist	₹3,000 – ₹15,000	Travel, journalists, field workers	"Protection in genuinely high-risk zones."

Architectural principle: Build only the BLE protocol and the Tier-1 reference device in-house. Every other SKU is an OEM/co-brand partnership. The phone is the source of truth; hardware is the trigger.

2. Tier 1 — Budget Trigger Devices

2.1 BLE Panic Keychain (Nordic nRF52810)

₹499

CR2032 · 12-month battery · BLE 5.0 · Custom GATT service

- Triple-press = SOS · single-press = silent check-in · long-press = fake call.
- Haptic motor confirms message-sent feedback.
- Pairs once with HearMe app via PSK + ECDH key exchange.
- BOM ~ ₹110; ample margin for D2C.

2.2 NFC Safe-Zone Tags

₹30 each

NTAG215 · Passive · No battery · Stick-on adhesive

Tap phone to mark "safely arrived" at home, hostel, office. Zero-UI check-in. Survives any hardware failure of the BLE button.

2.3 Stealth Pen Trigger

₹599

Concealed BLE module inside a ballpoint pen body

Cap-click triggers SOS. High discretion — passes any pocket-empty checkpoint and looks like a stationery item.

3. Tier 2 — Fashion Wearables

3.1 Smart Jewellery (D2C Co-Brand)

₹1,200–₹2,500

Pendant · Bracelet · Kada · Ring · Brooch

- Same nRF52 internals as Tier-1; premium gold/rose-gold finish.
- Double-tap = SOS; haptic ack on contact acknowledgement.
- Partner SKUs: **Melorra** · **GIVA** · **CaratLane** — "HearMe Edition."

3.2 Dupatta / Sari Pin (India-First Form Factor)

₹899

Micro BLE module disguised as a safety pin / brooch

Coercion-resistant: cannot be confiscated at checkpoints, looks like a traditional accessory. *No global safety app currently ships this form factor.*

3.3 Smart Hair Clip / Scrunchie

₹1,499

For students and college-age users

Haptic pattern signals back when a trusted contact has acknowledged the SOS — reassurance loop is felt, not seen.

3.4 Smart Insole / Shoe Module

₹2,200

Piezo-triggered foot-stomp pattern (three heavy stomps = SOS)

Hands-free trigger for situations where hands are restrained or holding things. Only worn-clothing trigger that survives a phone+jewellery confiscation.

4. Tier 3 — Specialist Hardware

4.1 HearMe Watch (ESP32-S3 + LTE-M)

₹6,500

- Own-brand smartwatch with eSIM — works even if phone is stolen.
- HR + SpO2 + skin-temp + GSR sensors fuel a physiological duress signature.
- Physical SOS slider on the crown (slide gestures are harder for an attacker to spot than taps).
- Cellular fallback when phone Bluetooth is lost.

4.2 Smart Helmet (2-Wheeler Riders)

₹4,500

Crash-detection + voice-keyword SOS + auto-ANPR record of the road behind. Co-brand with **Steelbird** or **Vega**.

4.3 AI Dashcam with Panic Pipe

₹7,900

OBD-II powered; on SOS, streams to HearMe cloud and runs ANPR on every passing vehicle. High value for cab drivers AND female passengers.

4.4 Satellite SOS Pendant

₹14,500

Iridium / Globalstar modem for genuinely dead zones — hikers, field NGO workers, rural school visitors. White-label via **Garmin inReach** or **Zoleo**.

5. Home & Environmental IoT

Smart Doorbell Integration

Ring / Realme / Mi doorbell webhook records 10 minutes pre & post SOS. Face-match flags strangers vs. trusted visitors.

Hidden-Camera Detector Dongle

USB-C / Lightning accessory with RF + IR-reflection scanner. One-tap scan of hotel rooms, Airbnbs, changing rooms. Massive organic word-of-mouth potential.

Smart Speaker De-Escalation

On home-zone SOS, Alexa / Google Home loudly plays a male voice: *"Who's at the door? I'm coming down."* Creates illusion of presence.

Drug-Test Coaster / Nail Polish

Co-brand with SipChip / Undercover Colors: detects GHB & Rohypnol. BLE variant logs result + location to evidence locker.

Smart Bulb Red-Strobe Alert

Philips Hue / Wipro bulbs flash red on SOS — visible signal to neighbours from outside the building.

Indoor Panic Pendant (Elder-Care Cross-Over)

Tata / Jio-style pendant with a Zigbee hub. Same app serves the elderly parent AND the adult daughter.

6. Vehicle & Mobility IoT

6.1 OBD-II HearMe Dongle

₹1,500

Plugs into any car > 2008. Detects airbag deploy, hard braking, route deviation. Works regardless of driver's phone state.

6.2 Uber / Ola / Rapido BLE Beacon

Partner program: registered cabs carry a BLE beacon tied to the verified driver's MAC. HearMe app auto-logs which driver you were with — chain of custody for any later complaint.

6.3 Smart Seatbelt Clip

Detects unbuckling during an active trip (likely forced exit) — silent SOS.

6.4 E-Scooter / E-Bike Fall Detection

Pair with Yulu / Bounce / Ola Electric: accelerometer + tilt = crash detection; auto-call emergency.

7. Detection & Active-Deterrent Hardware

Device	Function	Indicative Price
130 dB Personal Alarm Keychain	Pull-pin siren + strobe + BLE-paired silent SOS. Loudness is itself the deterrent.	₹699
IoT Pepper-Spray Cap	Removing safety pin triggers SOS + audio + GPS upload. Self-defence is logged at the moment of use.	₹1,200
Ultrasonic Attacker Deterrent	110+ dB ultrasound bursts — inaudible to humans, disorients certain attackers and dogs. Experimental.	₹1,800
Thermal-Imaging Phone Clip	FLIR One module: detects a hidden person behind curtains, bushes, inside a car boot. Hotel and parking-lot safety routine.	₹8,500

8. Biometric Wearable Layer

Signal	Sensor	Safety Use
Heart-rate spike with no motion	PPG	Fear response while sitting still — a strong silent-distress signal.
GSR (skin conductance)	GSR electrodes	Stress biomarker; reliable during panic events.
Skin temperature drop	IR thermopile	Shock / freezing response.
Blood oxygen	SpO2	Crash or unconsciousness fallback trigger.
Wrist tension (sEMG)	Whoop-style sEMG	Fight-or-flight arousal.

Fusion rule: any two of the five physiological signals firing together is a "physiological duress signature." Combined with the existing behavioural biometrics in HearMe, this becomes a world-class trust engine that no current safety app can match.

9. Network Infrastructure IoT

9.1 LoRaWAN Panic Beacons (Campus / Factory)

10 km range, 5-year battery, ₹800 per unit. Deploy free to colleges with one HearMe gateway on the admin block. No SIM needed. Classic B2B play — target 500 universities.

9.2 NB-IoT Mini-Tracker

Cheaper than smartwatch cellular; "bag tag" for women commuting alone. Works without the phone at all. Use Airtel / Jio IoT SIM.

9.3 Bluetooth-Mesh Canopy (Metro Stations)

Deploy HearMe BLE gateways across Delhi Metro / Bengaluru Metro stations. Any user in distress on a platform routes SOS through infrastructure. Strong PPP / grant opportunity with transit authorities.

10. B2B Institutional Deployment Kits

HearMe Campus Kit

CSR Pitch

100 × T1 panic buttons + 10 × LoRa gateways + admin dashboard. Indian corporates must spend 2% profits on CSR — ready-made target.

HearMe Transit Kit

State Tender

In-bus panic button at every seat + DVR + HearMe backend. Delhi, Bengaluru, Hyderabad have active women-safe-bus tenders.

HearMe Workplace Kit

PoSH Act

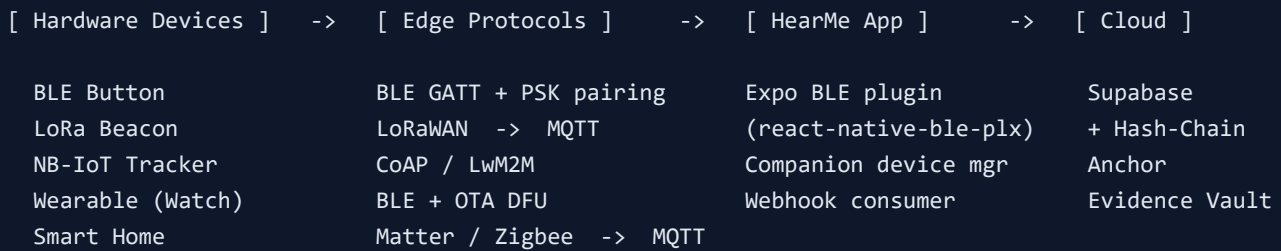
Desk-fob button + admin duty-of-care dashboard. IT and Facilities will pay for compliance with PoSH Act 2013 obligations.

HearMe NGO Kit

NGO Bundle

500 T1 buttons + offline BLE-mesh gateway + multilingual training material. Suitable for rural deployment via Sakhi / women SHGs.

11. Reference Technical Architecture



Security Posture (non-negotiable)

- All devices use pre-shared-key + ECDH pairing. No "enter 000000" weak PINs.
- Firmware OTA signed with HearMe ed25519 key.
- Every GATT write carries a replay-protected monotonic counter.
- Devices store nothing identifying — the phone is the only source of truth.
- Cloud-side hash-chain anchoring ensures evidence is court-admissible.

12. Partnership Map — Don't Build, Integrate

Category	Don't Build	Partner With
Smartwatch	Your own SKU	Noise · boAt · Amazfit co-branded edition
Jewellery	Your own line	Melorra · GIVA · CaratLane
Dashcam	Custom hardware	70mai · DDPAI (Indian presence)
Satellite SOS	From scratch	Garmin inReach · Zoleo white-label
Home sensors	Custom	Mi Home · Realme Link · Matter
Helmet	Your own	Steelbird · Vega OEM MOU
Pepper spray	Custom canister	SABRE India with HearMe pin module

Build only **Tier-1 panic buttons** and **BLE gateway firmware** in-house. Everything else stacks on top once the protocol is stable.

13. Recommended First Hardware Ship

The HearMe Clip — ₹499

A single SKU: a BLE panic keychain with pin and magnetic back so it can clip inside a bag, on a bra strap, on a keyring, or on a school ID lanyard. Runs on a CR2032 cell for a year.

Why this one device first:

- Hits the Tier-1 mass market price point.
- Lets you validate the BLE pairing UX inside the existing HearMe app.
- Becomes the reference device every future partner SKU must interoperate with.
- Easy to bundle into NGO and university CSR grants.

Everything else — jewellery, watches, helmets, dashcams — stacks on top once this protocol is stable in production.

Appendix — 12-Month Hardware Roadmap

Quarter	Milestone	Outcome
Q1	HearMe Clip prototype + BLE protocol v1 + app pairing UX	10 internal testers; firmware OTA pipeline live
Q2	Pilot launch via Amazon & D2C site; first NGO partnership	2,000 units shipped; Trust-Engine + hardware data fused
Q3	Jewellery co-brand SKU 1 (pendant); 3 university campus deployments	First B2B revenue; LoRa gateway firmware v1
Q4	HearMe Watch alpha; OBD-II dongle pilot with one ride-share partner	Cellular-fallback architecture proven; PR moment